Arista Access Node Optical Transport Failure Resolution SOP (BAS-ACC-62)

Node: BAS-ACC-62 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_8201

Description

The Access Node (AN) plays a crucial role in the Arista network architecture as the initial point of entry for customer traffic. The Optical Transport Failure (OTF) on BAS-ACC-62, Site Location: Basavanagudi POP, is primarily attributed to the degradation of SFP/XFP modules or fiber cuts, leading to Rx/Tx power loss. This failure domain occurs due to the inherent limitations of optical transmission technology, where signal attenuation and noise are inevitable over long-distance fiber links. The Arista AN's role in this context is to detect and report OTF events through alarms (e.g., ALARM_8201), triggering the need for swift resolution to maintain network uptime and Quality of Experience (QoE). The underlying protocol mechanics responsible for this failure domain include the Ethernet OAM (Operations, Administration, and Maintenance) protocol, which relies on optical signals for network monitoring and management.

Pre-requisite Checks

- item: Verify the site's environmental conditions (temperature, humidity, and power supply) are within the recommended operating parameters., description: Ensure the Basavanagudi POP's climate-controlled environment is functioning correctly to prevent damage to the equipment.

- item: Check the redundant link status on BAS-ACC-62., description: Verify the Standby link is operational and ready to take over in case of a primary link failure.

- item: Gather information from the previous alarm history (if any) to understand the failure pattern., description: Review the system logs to identify any recurring issues or correlated events that may be contributing to the OTF.

Potential Root Causes

Cause	Probability
Degradation of SFP/XFP module	High
Fiber cuts or damage	Medium
Rx/Tx power loss due to signal attenuation or noise	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface tengigabitethernet 1/1/1, Expected Output: Interface: tengigabitethernet 1/1/1, Status: up, Link Length: 100m, Rx Power: -10 dBm, Tx Power: -5 dBm, Description: This command displays the interface status, link length, and power levels for the specified interface. It helps identify if there are any issues with the SFP/XFP module or fiber connection.
```

```
Command: show alarm summary, Expected Output: ALARM_8201 - OTF
detected on interface tengigabitethernet 1/1/1, Description: This
command provides a summary of all active alarms on the system,
including the OTF event triggered by the ALARM_8201 code.

Command: show ethernet oam status, Expected Output: OAM status:
enabled, Loopback status: disabled, Description: This command displays
the Ethernet OAM status and loopback configuration. It helps verify if
the OAM protocol is functioning correctly and if the loopback is
enabled, which can affect the OTF detection.
```

Resolution Steps

Step 1: Step: Replace the faulty SFP/XFP module (if degraded), Command: install transceiver tengigabitethernet 1/1/1 new-sfp, Description: This step involves replacing the degraded SFP/XFP module with a new one to ensure reliable optical transmission.

Step 2: Step: Verify the fiber connection and repair any damage (if detected), Command: show interface tengigabitethernet 1/1/1 detail, Description: This step requires inspecting the fiber connection and repairing any damage to ensure a secure and reliable link.

Step 3: Step: Adjust the Rx/Tx power levels (if necessary), Command: set interface tengigabitethernet 1/1/1 rx-power -10 dBm, Description: This step involves adjusting the Rx/Tx power levels to compensate for signal attenuation or noise, ensuring optimal signal quality.

Post-Check Verification

```
Command: show interface tengigabitethernet 1/1/1, Expected Output:
Interface: tengigabitethernet 1/1/1, Status: up, Link Length: 100m, Rx
Power: -10 dBm, Tx Power: -5 dBm, Description: This command is used to
verify that the interface is operational and the SFP/XFP module is
functioning correctly.

Command: show alarm summary, Expected Output: No active alarms,
Description: This command is used to verify that the OTF event has
been cleared and there are no other active alarms on the system.

Command: show ethernet oam status, Expected Output: OAM status:
enabled, Loopback status: disabled, Description: This command is used
to verify that the Ethernet OAM protocol is functioning correctly and
the loopback is disabled.
```

Roll-back Procedures

- Step: Restore the previous SFP/XFP module (if replaced), Command: install transceiver tengigabitethernet 1/1/1 old-sfp, Description: This step involves restoring the previous SFP/XFP module to ensure the system returns to its original state.

- Step: Re-adjust the Rx/Tx power levels (if adjusted), Command: set interface tengigabitethernet 1/1/1 rx-power original-value, Description: This step involves re-adjusting the Rx/Tx power levels to their original values to ensure the system returns to its original state.